Project Design Phase
Proposed Solution Template

Date	9th march 2026
Team ID	EL2026TMID10104
Project Name	AI-Based Anomaly Detection Platform
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Modern healthcare systems generate a large amount of patient monitoring data through wearable devices and medical sensors such as heart rate monitors, blood pressure sensors, and oxygen saturation trackers. However, most existing systems analyze this data in batches rather than in real time. Due to this limitation, abnormal health conditions such as sudden heart rate spikes, irregular blood pressure levels, or oxygen drops may not be detected immediately, which can delay medical response and increase health

		risks for patients. Therefore, there is a need for a **real-time intelligent monitoring system** that can continuously analyze patient data streams and instantly detect abnormal patterns so that timely alerts can be generated for healthcare professionals.
2.	Idea / Solution description	The proposed solution is an **AI-driven real-time healthcare monitoring system** that detects anomalies in patient health data using machine learning and streaming technologies. The system collects continuous patient health data from sensors or simulated devices and streams it using **Apache Kafka**. The data is then processed by a machine learning model trained using anomaly detection algorithms such as **Isolation Forest and Autoencoder**. When the model identifies abnormal patterns in the incoming data, the system immediately generates alerts. These alerts are sent to healthcare providers through **email notifications** and displayed on a **real-time monitoring dashboard**. The system is built using the following technology stack: • **Python** for data processing and

		machine learning • **Apache Kafka** for real-time data streaming • **Flask** for backend API services • **PostgreSQL** for storing patient data and anomalies • **Machine Learning models** for anomaly detection • **Web dashboard** for visualization and monitoring This architecture enables continuous monitoring and quick detection of potential health risks.
3.	Novelty / Uniqueness	The uniqueness of this project lies in the integration of **real-time streaming technology with machine learning-based anomaly detection** in the healthcare domain. Unlike traditional healthcare monitoring systems that analyze historical data, this system: • Processes **live streaming data** • Uses **unsupervised machine learning models** • Detects anomalies **instantly** • Generates **automatic alerts for abnormal health conditions**

		The combination of **Kafka streaming, ML models, and real-time dashboards** provides a scalable and intelligent monitoring solution.
4.	Social Impact / Customer Satisfaction	This project can significantly improve healthcare monitoring by enabling **early detection of abnormal health conditions**. Benefits include: • Faster medical response during emergencies • Continuous monitoring of patients in hospitals or at home • Reduced workload for healthcare professionals • Improved patient safety and health outcomes • Support for remote healthcare and telemedicine By providing real-time insights and alerts, the system helps healthcare providers make **faster and more informed decisions**, leading to improved patient satisfaction and better medical care.

5.	Business Model (Revenue Model)	The proposed system can be implemented as a **Software-as-a-Service (SaaS) healthcare monitoring platform**. Possible revenue models include: • Subscription-based services for hospitals and clinics • Licensing the software to healthcare institutions • Integration with wearable device manufacturers • Cloud-based monitoring services for remote healthcare Hospitals and healthcare providers can pay for **real-time patient monitoring solutions** that improve operational efficiency and patient care.
6.	Scalability of the Solution	The system is designed to be highly scalable due to the use of **Apache Kafka and modular architecture**. Key scalability features include: • Kafka allows handling of **high-volume streaming data from thousands of devices** • Microservice-based architecture enables independent scaling of components • Cloud deployment allows flexible

		resource management • Machine learning models can be retrained and improved with more data This makes the solution suitable for deployment in **large hospitals, healthcare networks, and remote patient monitoring systems**.